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First paleomagnetic data from the sedimentary cover of the French Penninic Alps: evidence for Tertiary counterclockwise rotations in the Western Alps

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Abstract

We present a paleomagnetic study performed in the Briançonnais sedimentary cover of the western Alpine Arc (France). Sampling was focused on Upper Jurassic rocks of the Briançonnais cover in the Briançon–Guillestre area. More than 100 samples from 11 sites were collected essentially from Ammonitico rosso limestones and sedimentary dykes of the Upper Jurassic. Thermal and alternating field demagnetizations revealed three components of magnetization carried dominantly by magnetite: (1) a low-temperature component with present-day magnetic field direction of viscous origin; (2) a well defined reverse-polarity component with a maximum unblocking temperature of 440ºC (A component); and (3) a weak high-temperature component (B component) that experienced a viscous overprint during heating and was difficult to isolate. The A component exhibits a negative fold test and shows a better clustering when corrected for tilting associated with late Alpine extension. This component is interpreted as a secondary remagnetization acquired during the cooling path related to Late Eocene–Early Oligocene metamorphism. The mean A component direction, corrected for late Alpine extensional tilting, is $D = 142^{o}$, $I = -57^{o}$, $k = 44$, $\alpha_{95} = 8^{o}$. This direction significantly differs by $47^{o} \pm 13^{o}$ from the expected direction for Eurasia. We interpret this difference to be related to a post-Eocene counterclockwise rotation about a vertical axis of the Penninic zone relative to stable Europe. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Western Alps; rotation; paleomagnetism; remagnetization

1. Introduction

In the Western Alps (Fig. 1), most kinematic models have considered that since the Late Cretaceous, the convergence between Africa and Eurasia has been dominantly accommodated by both thickening and horizontal translation of tectonic units between major strike-slip faults [1–3]. According to these models, rotations about vertical axes are a local-scale mechanism [1,4]. Large-scale rotations involving most of the units of the western Alpine Arc have, however, been suggested on the basis of regional fault and strain patterns [5–7]. According to Ménard [6], the southwestward extrusion of the Western Alps associated with the indentation

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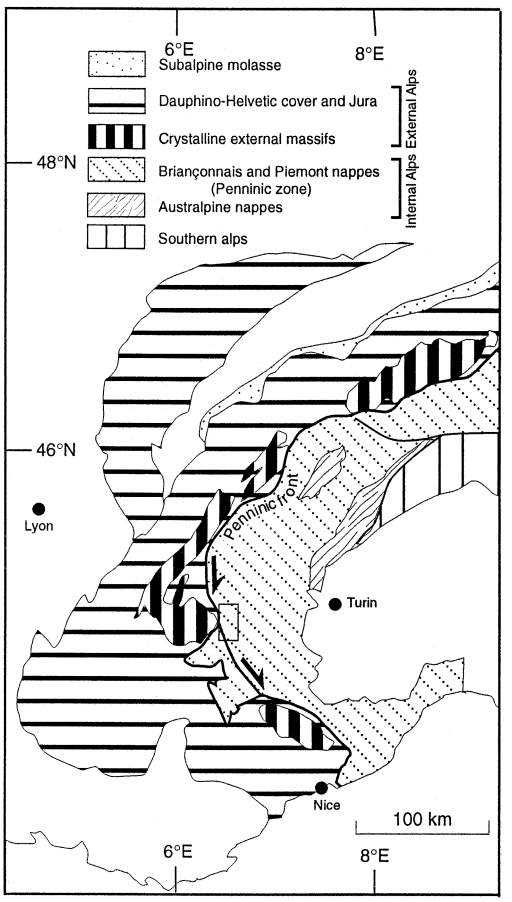


Fig. 1. The main tectonic units of the Alpine Arc. The thicker line marks the boundary between the internal and external Alps. Box indicates the studied area.

of Adria into Europe is accommodated, since the Eocene, by translation and counterclockwise rotation up to 43° along curved longitudinal faults (e.g. the Penninic front) (Fig. 1). Paleomagnetism is a powerful method for measuring rotations subsequent to magnetization acquisition but very few data on Cenozoic rocks are available for the internal part of the Western Alps [8] where the largest rotations are expected [6]. Such a lack of paleomagnetic data in the Western Alps is related to the scarcity of sedimentary facies suitable for a paleomagnetic study, especially in the internal zone, and from the fact that paleomagnetic surveys in the Western Alps were mainly carried out when high-sensitive cryogenic magnetometers were not yet available. In this framework, we have conducted an analysis of Jurassic rocks of the Briançonnais zone in the French Alps to test (1) the suitability of these rocks for paleomagnetic analysis and (2) the various kinematic models for the Western Alps during the Tertiary. We have focused our study on the domain located between the cities of Briançon and Guillestre (Figs. 1 and 2) where large- and small-scale structures are accurately constrained by field data [9–11]. This area is the least deformed and metamorphosed part of the Briançonnais zone. During Late Eocene–Early Oligocene deformation, high-pressure–low-temperature metamorphism occurred in the Briançonnais zone, leading at least to a partial remagnetization of natural remanent magnetization (NRM). We show that Mesozoic rocks of the Briançonnais zone between Guillestre and Briançon carry a well-defined post-tectonic partial remagnetization related to Late Eocene–Early Oligocene alpine metamorphism.

2. Geological overview of the study area

The Briançonnais zone is part of the Penninic Alps (Fig. 1) and consists of rocks deformed and metamorphosed during the Tertiary. This domain was initially part of the European passive margin of the Ligurian Tethys that separated Eurasia from the Adria plate during the Jurassic and Early Cretaceous [12].

In the Western Alps, the Briançonnais zone consists of two main stratigraphic units: (1) a pre-Mesozoic basement made of crystalline rocks overlain by Permo–Carboniferous continental sediments; and (2) a Mesozoic cover. The Mesozoic sequence consists of Triassic to Early Jurassic platform carbonates with intercalations of evaporites, overlain by late Dogger neritic carbonates and Late Jurassic to Paleocene pelagic sediments. The Late Jurassic sequence consists of massive pelagic limestone beds, up to 30 m thick, with a basal layer of nodular limestone in a pale red carbonaceous matrix (Ammonitico rosso facies). A few levels of red crinoid limestone also occur. Several sedimentary dykes (neptunian dykes) associated with Upper Jurassic tectonic mobility were synchronously filled with pelagic sediments similar to Ammonitico rosso [13]. At the top of the Upper Jurassic, a thin (up to several centimeters thick) red

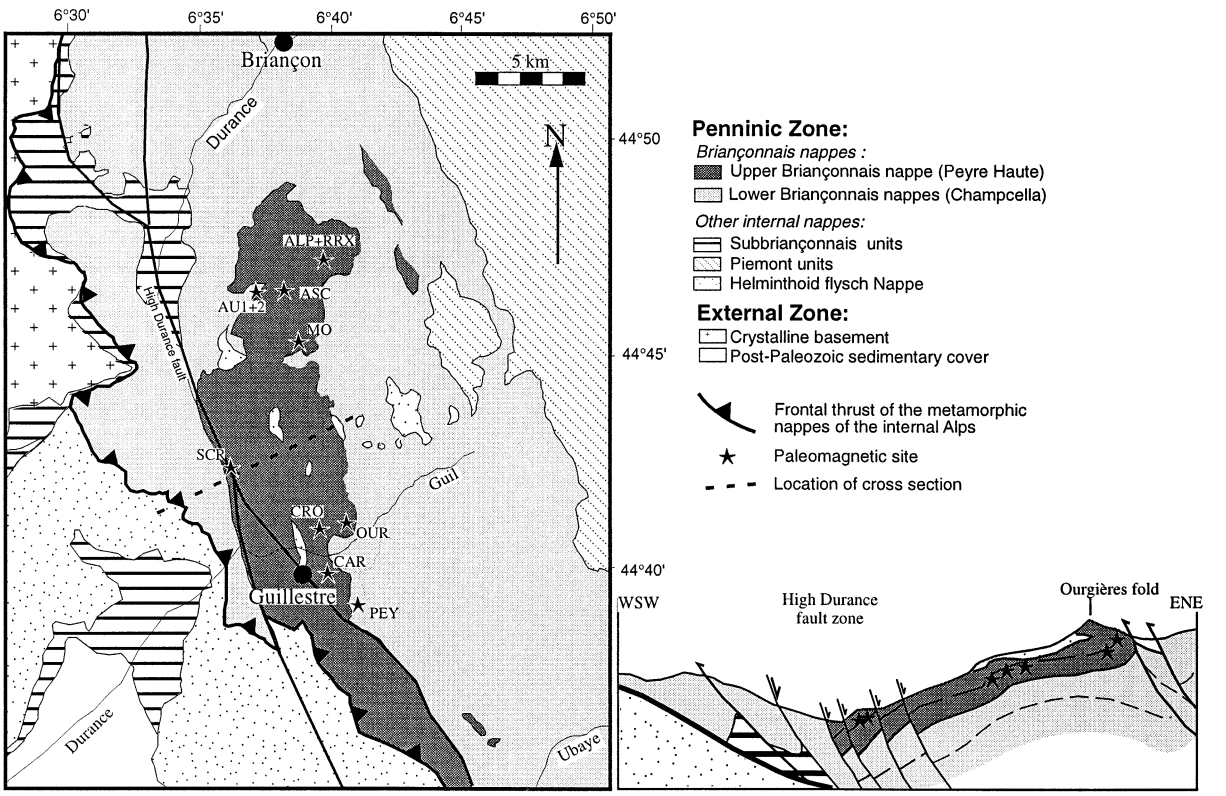


Fig. 2. Structural map and simplified cross-section of the Briançonnais zone in the Durancian segment. The paleomagnetic sites are projected on the cross-section. Normal faults associated with late Alpine extension bound tilted blocks.

or green condensed level (hard-ground) corresponds to the Early Cretaceous and is overlain by thick Upper Cretaceous calcschists. The Upper Jurassic and hard-ground deposits were the main target for this paleomagnetic study.

The structure of the Briançonnais zone is a stack of nappes essentially made of the Meso–Cenozoic sequence. The present-day structure of the Briançonnais zone is the result of several deformation phases. The first phase (D1) involved the emplacement of Briançonnais nappes associated with a HP–LT metamorphic event during the Late Eocene. Lawsonite–Albite facies are described in the Houillère zone of the Vanoise massif [14] and large-scale isoclinal folding occurred in the study area (e.g. Ourgière fold, Fig. 2). The second phase (D2) involved refolding of the Briançonnais nappes and a general thrusting over the external domain accompanied by Prehnite–Pumpellyite facies HP–LT metamorphism. This was followed by a third (D3), non-metamorphic, east-verging folding event [9]. The youngest sediments involved in the D2 deformation are locally dated as Early Oligocene but this age is poorly constrained, as is the D3 deformation event. Finally, late Alpine extension and dextral strike-slip displacement along longitudinal faults bounding tilted blocks occurred in the Durancian segment of the Briançonnais zone (Fig. 2) during the late Cenozoic [10,11,15,16].

3. Paleomagnetic sampling and measurement

More than 100 samples from 11 sites were collected in Upper Jurassic rocks of the Briançonnais cover (Table 1; Fig. 2). Six sites were collected in the Ammonitico rosso, four sites in the sedimentary dykes, one site in the hard-ground at the top of the Late Jurassic, and a few samples of a site (AU1) in red crinoid limestone. In order to test differential rotations between thrust sheets, we chose,

Table 1
Data on paleomagnetic sites

Site	Latitude (°N)	Longitude (°E)	Nappe	Rock	S_0 dd/dip	S_1 dd/dip
ALP	44.803	6.66	Upper	SD	359/125	
RRX	44.803	6.66	Upper	Am. rosso	horizontal	
ASC	44.786	6.64	Upper	Hard-ground	220/10	
AU1	44.783	6.621	Lower	SD + calc		
AU2	44.783	6.621	Lower	SD	(1) 73/36; (2) 128/55	
OUR	44.685	6.641	Upper	Am. rosso		358/30
GRO	44.685	6.633	Upper	Am. rosso	(1) 240/15; (2) 168/75	342/58
SCR	44.706	6.593	Upper	Am. rosso		295/26
CAR	44.065	6.658	Upper	Am. rosso	(1) 240/15; (2) 190/39	216/34
PEY	44.646	6.658	Upper	Am. rosso	(1) 235/35; (2) 235/20	215/50
MO	44.783	6.624	Upper	SD	153/83	

Nappe: Upper = upper Briançonnais nappe, Lower = lower Briançonnais nappes. Rock: SD = sedimentary dyke, Am. rosso = Ammonitico rosso, Calc = crinoid limestone. S_0/S_1: bedding/foliation, dd = dip direction, dip = inclination. For sites located on tilted blocks, we indicate bedding related to (1) late Alpine extensional tilting and (2) to earlier folding. Bedding is not reported when it varies between different cores at one site.

when possible, sites in the two main superimposed nappes of the area: the Champcella and Peyre Haute nappes (Figs. 2 and 3; Table 1). Specimens were collected in the field using portable gasoline-powered equipment and oriented with a magnetic compass. Natural remanent magnetization was measured with a two-axis cryogenic magnetometer (LETI). Thermal demagnetizations were done with a Schonstedt TSD1 furnace and alternating field demagnetizations were done with a Schonstedt AF demagnetizer.

4. Magnetic behavior of the lithologies

Intensity of the NRM varies between 1×10^{-4} A m^{-1} and 1×10^{-3} A m^{-1}. However, most of the samples show low-intensity NRM between 1×10^{-4} A m^{-1} and 3×10^{-4} A m^{-1}. One site located on a mountain peak (RRX) revealed high NRM intensities (1×10^{-2} A m^{-1}) carried by low-coercivity magnetic minerals and most probably corresponding to a natural isothermal remanent magnetization. This site is not considered in the following discussion.

4.1. Thermal demagnetization of the NRM

Within most of the sites, three components of magnetization were observed (Fig. 3a–d): (1) a low unblocking temperature component (maximum 160–230°C) showing a present-day magnetic field direction; (2) a second reversed component (A), responsible for the major part of the NRM has been found in all specimens with a maximum unblocking temperature between 390°C and 440°C; and (3) a weak high-temperature component (B) with an unblocking temperature above 450°C. This component has a low intensity and was difficult to isolate. Indeed, above 450°C, a strong increase in bulk susceptibility was observed simultaneously with the appearance of a viscous magnetization (Fig. 4a). In addition, demagnetization paths on stereographic projection often occur along great circles which suggests a partial overprint of the B component by viscous magnetization (Fig. 4b). We attribute this behavior to the growth of new magnetite at a temperature of 450°C [17]. New grains are very small and have very short magnetization relaxation times. The maximum unblocking temperature of the B component was therefore difficult to determine, although it can be estimated to reach about 600°C.

4.2. Combined alternating field and thermal demagnetization of the NRM

On several pilot samples, alternating field (AF) and thermal demagnetization (Fig. 3e) were combined. First the sample was submitted to AF up to 0.1 T, followed by a thermal demagnetization up to

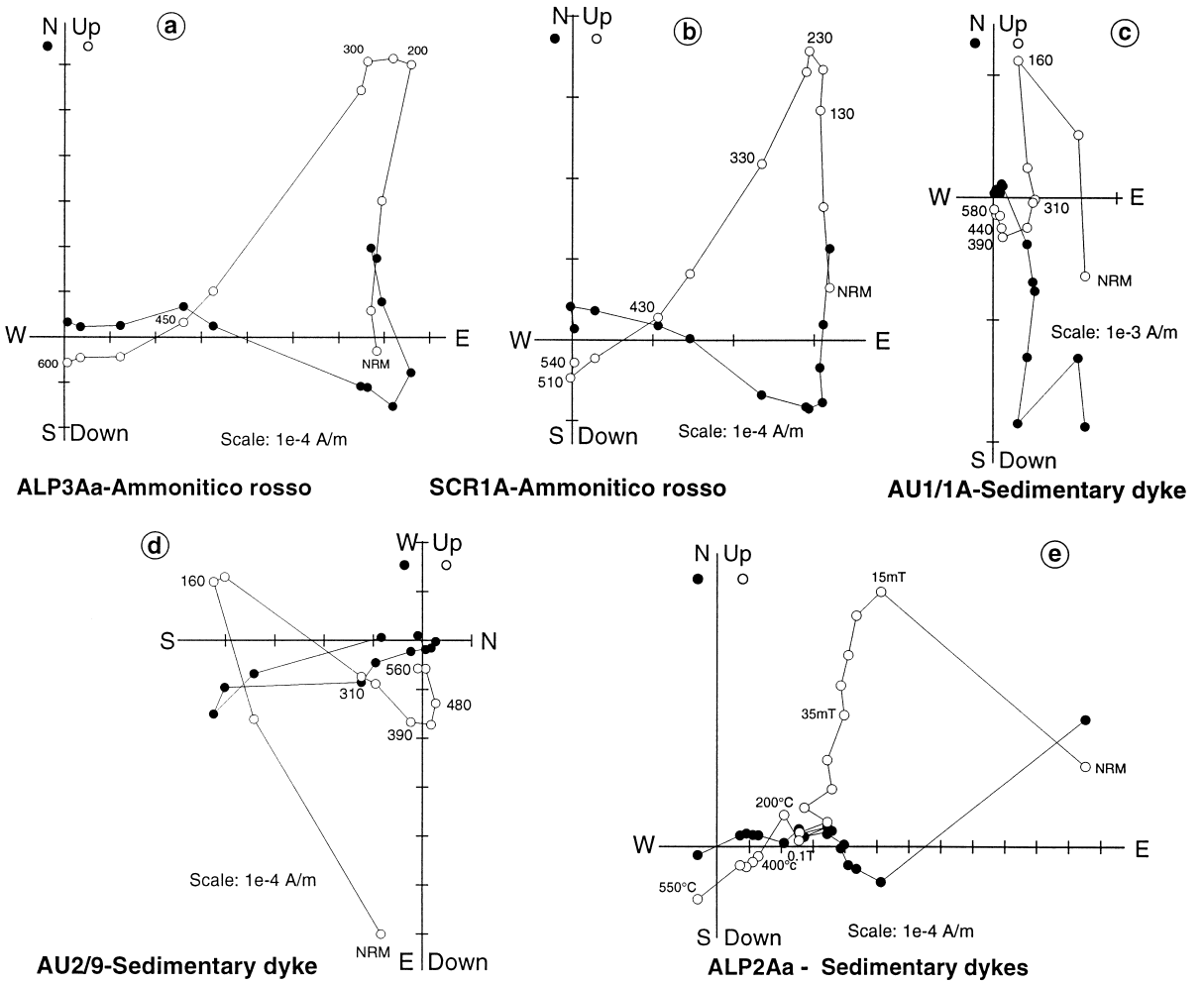


Fig. 3. Orthogonal plots in geographic coordinates of representative demagnetizations. Circle, vertical component; dot, horizontal component. (a–d) Thermal treatment with numbers indicating temperature step. A low-temperature component with maximum unblocking temperature of 230ºC is observed with a present-day magnetic field direction. The A component shows maximum unblocking temperature between 390ºC and 440ºC and is always of reverse polarity. The B component, with unblocking temperature above 450ºC is often observed but difficult to isolate. (e) Combined alternating field and thermal treatment. The low-coercivity component is observed but poorly defined. A component, close to the A component observed with thermal demagnetization, is removed with a median destructive field between 15 mT and 70 mT. This component does not converge to the origin. Thermal demagnetization following alternating field demagnetization is inefficient to isolate the third component of higher coercivity.

550ºC. After removing a low-coercivity component at 15 mT, a second component, close to the A component (revealed with thermal treatment) is observed, with a medium destructive field between 15 mT and 70 mT. Approximately 70% of the NRM is removed at 0.1 T, and in most cases demagnetization paths do not converge to the origin, indicating a third component of higher coercivity. Thermal demagnetization partially removes the remaining NRM, but partial viscous remagnetizations similar to those observed with thermal treatment were observed above 450ºC.

From demagnetizations, we conclude that the NRM is composed of three magnetization components. Thermal treatment was the most efficient way to separate the low temperature (low coercivity) and the A components (intermediate tempera-

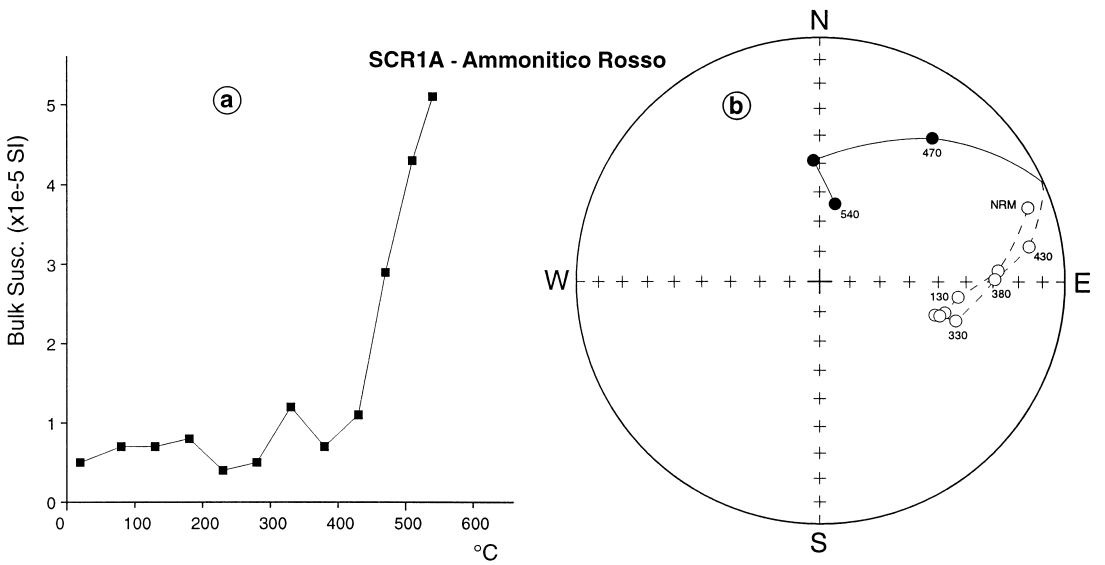


Fig. 4. Representative behavior of the NRM during heating. (a) Evolution of susceptibility during thermal demagnetization. A strong increase of susceptibility is observed above 450ºC. (b) Stereographic projection of thermal demagnetization (geographic coordinates); dot, normal polarity; circle, reverse polarity. Demagnetization path occurs along a great circle above 450ºC, showing a partial overprint of the B component by viscous magnetization.

ture, medium coercivity). The B component (above 450ºC, high coercivity) is poorly defined. Lastly, a majority of the NRM is carried by a low- to medium-coercivity phase.

4.3. IRM acquisition and thermal demagnetization of three IRM components

Representative samples were analyzed by progressive acquisition of isothermal remanent magnetization (IRM) (Fig. 5a–c) followed by thermal demagnetization of three mutually orthogonal isothermal remanence components acquired at a field of 1.0, 0.4 and 0.1 T, respectively [18] (Fig. 5d–f). The IRM acquisition curves show a variable contribution of coercivity phases. Most of the samples do not saturate at 1.0 T, which indicates a fraction of high-coercivity magnetic minerals. The fraction of low-coercivity magnetic minerals differs significantly between the lithologies. Thermal demagnetizations of three IRM components (Fig. 5d–f) show that the low-coercivity phase (0.1 and 0.4 T) dominates in most cases (Fig. 5d,f) and 90% of the IRM is removed below 600ºC. After this first thermal cleaning, a second acquisition of three orthogonal IRMs was performed followed by thermal demagnetization. In all cases, intensity of IRM is significantly higher than that for the first procedure (described above) and is carried by a low-coercivity phase, indicating growth of new magnetite during thermal treatment (also suggested by the increase of the low field susceptibility above 450ºC). Furthermore, the high-coercivity component with unblocking temperature around 450–500ºC (Fig. 5d) is not stable during heating as it is not observed during the second thermal treatment. This component could be carried by maghemite.

From these tests, we conclude that magnetite is the main magnetic carrier for the low-coercivity phase, and therefore for most of the NRM. Maghemite is also observed as well as pigmentary hematite, responsible for the red colors of the Upper Jurassic limestones and sedimentary dykes, but as minor ferromagnetic phases.

5. Paleomagnetic directions

Analysis and comparison of paleomagnetic directions in geographic and stratigraphic coordinates

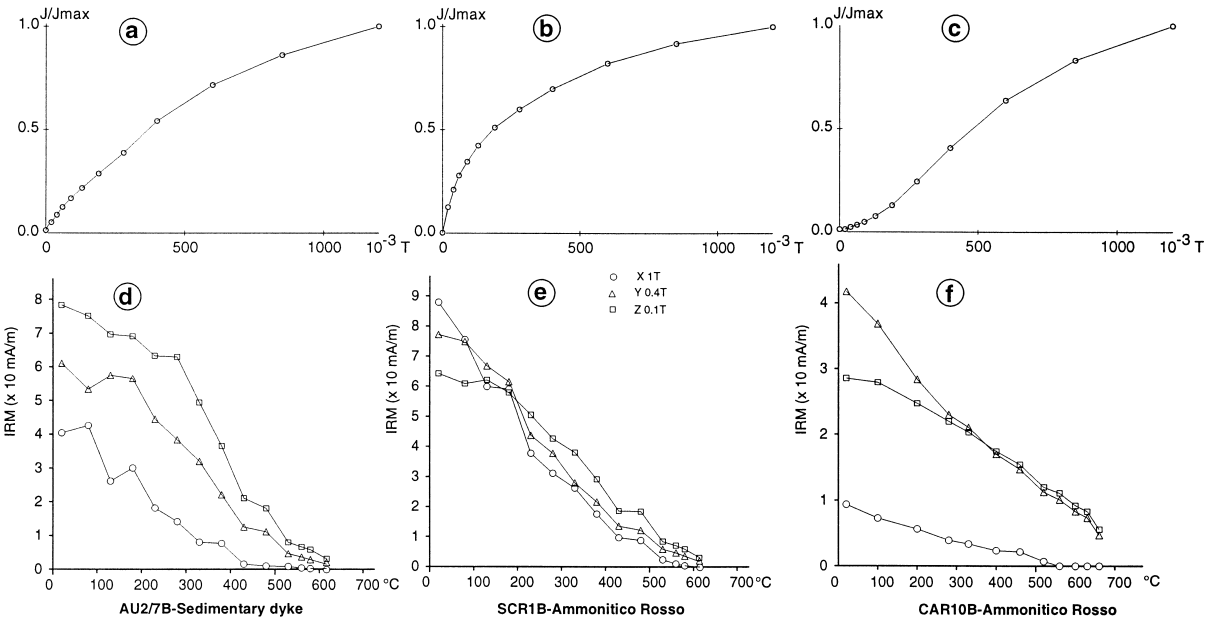


Fig. 5. (a) Isothermal remanent magnetization acquisition curves (IRM) and, (b) thermal demagnetization of three-axis IRM for representative samples. The IRM curves show a fraction of high-coercivity magnetic minerals, probably hematite. Thermal demagnetization of IRM shows that low- and intermediate-coercivity magnetic minerals, probably magnetite, dominate and carry most of the NRM.

require a knowledge of the deformation phases that occurred in the Briançonnais zone and their respective contribution to the present-day local bedding orientations. D1 and D2 deformation phases with associated metamorphism are responsible for most of the folding, while, in the study area, the D3 deformation phase has a minor contribution to the present-day folding [9]. Fold axes are mostly horizontal and oriented NW–SE. However, it was not possible to accurately separate in the field the respective contribution of the different compressive deformations. Late Alpine extension, probably of late Cenozoic age has been observed in the study area [10,11,15,16]. Normal faults oriented east-northeast and south-southwest accommodate tilting, up to 35º, of blocks hundred of meters to kilometers in scale (Table 1). Detailed field work has been done [10,11,15,16] to map the tilted blocks and associated normal faults. Extensional structures are clearly visible in the field at every scale as they vertically offset the earlier thrusts and folds [16]. Due to the different scales between extensional and earlier compressive structures, it was possible to identify sites located on tilted blocks and, for each of them, separate the respective contribution of the extensional tilting and earlier folding on present-day bedding orientation (Table 1).

The mean direction of the sites located on tilted blocks (AU1, AU2, GRO, CAR, PEY) were untilted in two successive steps: first, by removing tilting related to extension and, secondly, by removing tilting related to earlier folding (stratigraphic coordinates) (Tables 1–3). The remaining sites (ALP, ASC, OUR, SCR, MO) were untilted in one step by removing tilting related to folding. In order to test the age of each component relative to the deformation of the area, we performed two fold tests. A first test comparing site mean directions in geographic coordinates with those corrected for extensional tilting, and a second test comparing site mean directions corrected for extensional tilting with those in stratigraphic coordinates.

Directions of the low-temperature and A components were calculated using linear regression techniques and are reported in Tables 2 and 3. No reliable directions were obtained by linear and/or by remagnetization circles analysis on the B component and it will not be further discussed.

Table 2
Paleomagnetic data for the low-temperature component

Site	n/N	Geographic coor.				Corrected for late tilting				Stratigraphic coor.			
		D	I	k	α_{95}	D	I	k	α_{95}	D	I	k	α_{95}
ALP^{+}	10/14	5	68	72	5	5	68	72	5	296	−30	72	5
ASC^{+}	4/4	356	74	116	9	356	74	116	9	339	66	116	9
AU1*													
AU2	9/10	338	57	53	7	204	86	53	7	217	31	56	7
OUR^{+}	5/7	46	60	17	19	46	60	17	19	292	33	16	19
GRO	4/4	14	47	25	19	316	22	25	19	316	22	25	19
SCR^{+}	8/15	350	60	52	8	350	60	52	8	280	55	52	8
CAR	8/11	28	55	21	12	14	45	21	12	340	35	21	12
PEY	6/10	10	59	59	9	350	30	59	9	347	11	59	9
MO^{+}	5/9	31	59	40	12	31	59	40	12	338	−2	40	12
Mean	9/10	12	61	39	8	352	63	5	27	313	27	4	31

Corrected for late tilting = the first step of untilting and indicates mean directions after correction due to late Alpine extensional tilting. Crosses indicate sites where no correction for late Alpine tilting was applied (i.e. they are not located on tilted blocks) (cf. Table 1). Stratigraphic coordinates = the second step of untilting and indicates mean directions after correction due to earlier compressive deformations. n/N = number of samples/sites used for statistics (n) and number of samples/sites (N). Mean = mean global direction. D = declination; I = inclination; k = dispersion parameter; α_{95} = confidence circle at 95% confidence level. Star indicates site for which it was not possible to isolate stable components of magnetization.

Table 3
Paleomagnetic data for the A component

Site	n/N	Geographic coor.				Corrected for late tilting				Stratigraphic coor.			
		D	I	k	α_{95}	D	I	k	α_{95}	D	I	k	α_{95}
ALP^{+}	13/14	123	−64	70	5	123	−64	70	5	114	55	70	5
ASC^{+}	4/4	136	−54	12	27	136	−54	12	27	135	−44	12	27
AU1	9/12	168	−38	38	8	162	−56	25	10	162	−56	25	10
AU2	8/10	162	−33	43	8	161	−69	43	8	63	−44	22	12
OUR^{+}	6/7	165	−52	145	5	165	−52	145	5	111	−11	114	6
GRO	4/4	118	−78	51	13	135	−64	51	13	99	0	51	13
SCR^{+}	13/15	117	−51	50	6	117	−51	50	6	90	−29	43	7
CAR	11/11	142	−67	23	10	145	−52	23	10	127	−20	23	10
PEY	8/10	127	−79	56	8	140	−44	56	8	141	−24	56	8
MO^{+*}													
Mean	9/10	146	−59	17	13	142	−57	44	8	115	−23	4	28

Same symbols as Table 2.

In geographic coordinates the low-temperature component exhibits a mean direction $D = 12^{o}$, $I = 61^{o}$, $k = 39$, $\alpha_{95} = 8^{o}$. One site (AU1) had a poorly defined low-temperature magnetization component that did not allow computation of a mean direction. Scattering of mean directions strongly increases after each step of untilting, and fold tests [18] are negative at the 99% confidence level (Table 2). The low-temperature component is close to the present-day magnetic field and is probably a recent viscous overprint. This is substantiated by viscosity tests showing that low-temperature viscosity contributes to 20% to 40% of the NRM.

The A component was isolated in nine out of the ten sites. It exhibits a mean direction $D = 146^{o}$, $I = -59^{o}$, $k = 17$, $\alpha_{95} = 13^{o}$ in geographic coordinates (Table 3; Fig. 6a). After removing titling related to extension (Table 1), the mean direction is roughly similar but with a strong clustering of mean sites direction ($D = 142^{o}$, $I = -57^{o}$, $k = 44$,

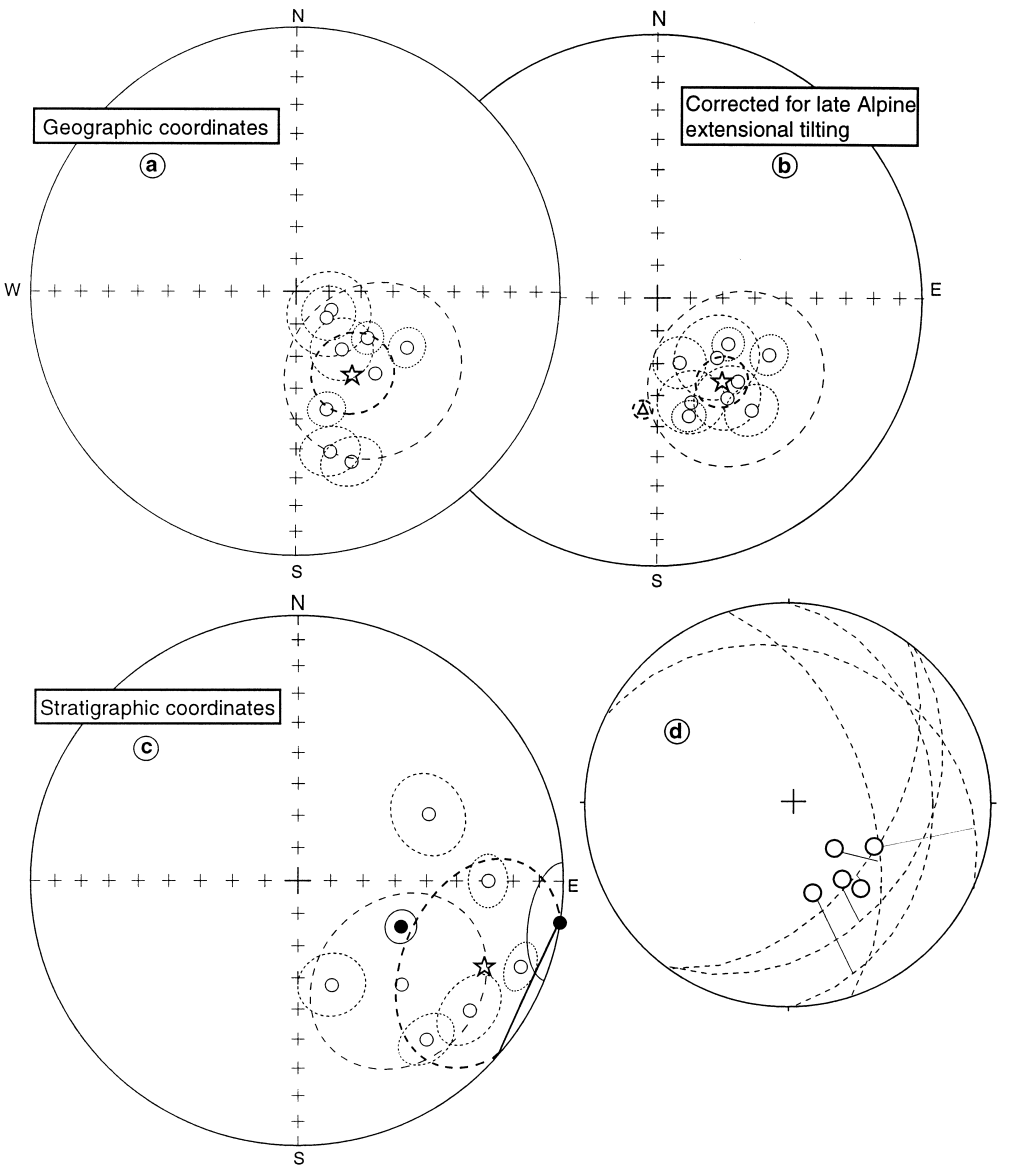


Fig. 6. Stereographic projection of the mean site directions of the A component with associated confidence circles. (a) Geographic coordinates. (b) Directions corrected for late Alpine tilting. Triangle is the reference direction for stable Europe at 30 Ma [31]. (c) Stratigraphic coordinates. Dot, normal polarity; circle, reverse polarity; star is the mean direction with associated confidence circle. (d) Tectonic foliation and mean paleomagnetic directions corrected for late Alpine tilting (connected by lines) for each of the Ammonitco rosso sites.

$\alpha_{95} = 8^{\circ}$) (Table 3; Fig. 6b). The fold test for this partial untilting is positive at the 99% confidence level. We therefore conclude that the A component was acquired prior to late Cenozoic extension. After removing the remaining tilt from the earlier folding phases, a strong scattering is observed and the mean direction becomes $D = 115^{\circ}$, $I = -23^{\circ}$, $k = 4$, $\alpha_{95} = 28^{\circ}$ in stratigraphic coordinates (Table 3; Fig. 6c). A fold test, comparing data in stratigraphic coordinates and data corrected for extensional titling is negative at the 99% confidence level. A significant decrease of the k parameter from geographic to stratigraphic co-

ordinates is also observed at the site level for the three sites (AU2, OUR, SCR) (Table 3) whose bedding dip is not uniform for all cores, in agreement with between-site negative fold test. Other paths of folding have been tested in order to consider, at some sites, uncertainty on extensional tilting (±5° at site CAR, ±10° at site PEY) and possible plunging of the fold axis (15° to the northwest at sites CAR and GRO). Small differences with the results presented above were observed for the *k* parameter at each step of untilting but fold test and mean directions were always very similar. This result suggests that the A component is a secondary magnetization acquired after the main folding phases of the area and prior to the late Alpine extensional tilting.

However, possible strain-rotation of remanence should be taken into account in the undeforming sequence (tilt and strain corrections) (see [19] for a review) as a penetrative deformation associated with D1 and D2 phases is observed throughout the study area. Detailed microstructural analysis has been done on the area [9] concluding that strain in the Ammonitico rosso limestones is moderate, dominantly oblate, with a foliation close to the bedding plane with no clear stretching lineation (X/Y ratio varies between 0.8 and 1.2). An apparent elongation within the bedding plane is visible, but this is due to an intersection between stratification and foliation. In addition, sedimentary dykes have suffered very weak strain compared to the Ammonitico rosso limestones. Hence, a moderate strain effect on a NRM acquired before deformation may only concern the inclination measured on Ammonitico rosso limestones. We consider this as a secondary effect in comparison with folding. Therefore, an untilting of the remanence direction as presented above is a sufficient test for the relative age of the remanence.

From fold tests, it was concluded above that the A component is a post-folding remagnetization. A deviation, due to rock anisotropy, of the A component during remagnetization, should nonetheless be considered. However, we do not suspect a significant deviation for the following reasons: first, directions for the weakly deformed sedimentary dykes are close to the observed directions for the stronger deformed Ammonitico rosso and, second, mean directions for the Ammonitico rosso are, for most of them, significantly deviated from foliation (Fig. 6d).

In conclusion, we infer (1) that the A component is a post-folding remagnetization acquired before late Alpine extensional tilting, and (2) that its direction is representative of the earth's magnetic field when it was acquired.

6. Absolute age and origin of the A component

The A component has the following main characteristics: (1) the maximum unblocking temperature is 440°C; (2) it is always of reverse polarity; (3) the fold test indicates that it is post-folding but prior to late Alpine extensional tilting. Most of the folding occurred synchronously with metamorphism during the two main deformation phases (D1 and D2) [9]. We therefore suggest that the A component is a remagnetization that occurred during post-metamorphic cooling of the buried Briançonnais units. K–Ar and Ar–Ar radiochronological data for the western internal Alps indicate ages older than 30 Ma [20]. In the absence of more accurate radiochronological data, we consider the remagnetization as Late Eocene–Early Oligocene. We may note, however, that remagnetization may have occurred prior to the last non-metamorphic deformation phase (D3). This compressive event had only a minor contribution to the bulk folding but could be responsible for the remaining scatter of data corrected for late Alpine extensional tilting (Fig. 6b).

At least two main mechanisms can be invoked to explain the remagnetization: (1) thermal activation of preexisting magnetic grains which acquired a thermoviscous remagnetization (TVRM); or (2) chemical remanent magnetization (CRM) acquired by the growth of authigenic magnetite assisted by tectonically driven fluids.

Theoretical models have been proposed for the time–temperature relationship of remanent magnetization in case of TVRM [21,22]. It is possible to calculate time and temperature conditions for TVRM knowing (1) the temperature that the rock has reached and (2) the length of time the rocks were exposed to this temperature. Maximum temperatures during the Oligocene metamorphic event reached 300–320°C in the external part of the Briançonnais zone [23–25]. Data on the Vanoise cover show that rocks remained between 270°C and 300°C for at

least a few million years after peak metamorphism [23]. Time–temperature diagrams [21] indicate that the TVRM acquired by magnetite grains, heated at 250ºC for 1 million years will be unblocked by heating to 400ºC for 1 h in a zero magnetic field. Hence, TVRM may explain unblocking temperatures for the A component between 400ºC and 450ºC. However, nomograms are valid for single domain magnetite, which may not be the case for the studied rocks. Furthermore, the A component only shows reverse polarity which suggests a fast remagnetization process while post-metamorphic cooling below unblocking temperatures occurred over several million years. In the Dauphinois zone located close to the study area, this process of remagnetization has been inferred by other authors but both polarities were observed [26].

Chemical remagnetization is a process that may occur during post-metamorphic cooling, especially when assisted by fluids. Large-scale fluid flow expelled from buried continental units during orogenic events are well known and may involve large volumes of fluids [27]. Fluid flow may react with local lithologies leading to dissolution of iron which then contributes to the precipitation of authigenic magnetite. In the Western Alps, evidence of important fluid flow associated with the last Alpine metamorphic event are inferred in the light of paleomagnetic data [28]. Most of the chemical remagnetization should have occurred in a short time span (maximum 10^6 yr), to explain the unique reverse polarity of the A component, although fluid flow events may have been pervasive for longer periods. Similar processes involving rapid chemical remagnetization have been inferred in other very low-grade metamorphic orogenic belts where a unique magnetic polarity is observed [29,30]. Therefore, a CRM is a possible alternative for remagnetization acquired in a short time during cooling associated with the Alpine tectono-metamorphic events. In this case, thermal magnetization processes may have also contributed to the remagnetization if tectonically driven fluids were hot. More detailed analysis, especially concerning magnetic mineralogy, would be necessary for a deeper understanding of the origin of the A component.

7. Discussion

The mean direction of the A component corrected for late Cenozoic extensional tilting is $D = 142^o$, $I = -57^o$, $k = 44$, $\alpha_{95} = 8^o$ (Table 3; Fig. 6b). The reference direction for stable Europe has been calculated using the apparent polar wander path (APWP) for Eurasia [31]. The APWP does not show substantial variations during Late Eocene–Oligocene time and therefore we use the reference at 30 Ma for comparison ($D = 189^o$, $I = -59^o$, $\alpha_{95} = 3^o$) (Fig. 6b). The paleoinclination of the A component is in good agreement with the chosen reference for a stable Europe. In contrast, the paleodeclination shows a large deviation of $47^o \pm 13^o$ [32,33] with respect to the reference direction. The deviation is systematic and significant for all sites. As discussed above, no significant internal effects (strain, anisotropy) deviating the magnetization are expected. Consequently, we consider that most of the observed declination difference is due to large-scale tectonic processes, i.e. tilting about a horizontal axis or rotation about a vertical axis. In the tilting hypothesis, a regional post-Eocene southeastward tilting of about 30º would be necessary to explain the discrepancy between measured and reference declinations. Structural evidence of tilting in the study area is associated with late Alpine extension but occurs at a smaller scale and involves in most cases a westward tilting along N–S normal faults dipping to the east (Fig. 1) [10,11,15,16]. Therefore, the tilting hypothesis is unlikely. We therefore infer that the deviation of the declination is related to counterclockwise vertical axis rotation of about 40º of the Penninic Alps relative to stable Europe.

Previous paleomagnetic data for the study area were obtained on Permian volcanics of the Guil Valley (Fig. 2) [34,35]. A well defined reverse-polarity component was isolated on both studies with a mean direction $D = 148^o$, $I = -33^o$, $k = 23$, $\alpha_{95} = 7^o$ in geographic coordinates, and $D = 152^o$, $I = -15^o$, $k = 25$, $\alpha_{95} = 7^o$ in stratigraphic coordinates [35]. Although this is an inconclusive fold test, the authors interpreted the component of magnetization as primary. However, the close similarity between this component in geographic coordinates and our A component rather suggests that a component of remagnetization related to Alpine metamorphism was

isolated, indicating, in turn, a same amount of rotation. Other paleomagnetic studies on Tertiary rocks of the internal Alps were performed on gneiss in the Lepontine dome area [36], on Oligo–Miocene intrusive rocks of the Traversella massif [37], and on volcanic rocks of the Sezia zone [38,39]. All studies suggest, for their respective areas, a Neogene counterclockwise rotation, in agreement with our data, but of lower amplitude (27º±13º [36], 25º [37] and 25º–30º [38,39]). This is probably due to the younger age of magnetization. However, the interpretation of paleodeclinations remained ambiguous because no paleohorizontal directions were available to properly untilt the data. In contrast, our data clearly show that the Briançonnais zone of the Western Alps has suffered, since the Late Eocene–Early Oligocene, a mean counterclockwise rotation of about 45º.

On the basis of kinematic reconstructions of the Penninic Alps, Ménard [6] has inferred that indentation of Adria into Europe was accommodated by a combined translation and counterclockwise rotations of the internal Alps. The author infers a total rotation of 43º since the Eocene, in agreement with our paleomagnetic data. The counterclockwise rotation of the Western Alps would be accommodated during the late Cenozoic by dextral strike-slip displacement along major longitudinal faults oriented N–S to NNW–SSE (e.g. the Penninic front, the High Durance fault and, to the south, the Serenne and Bersezio faults) (Figs. 1 and 2). We infer that these faults accommodate the counterclockwise rotation of the Penninic zone relative to stable Europe to the west. Considering a rotation axis fixed in the Po plain [6], a 40º rotation would be responsible for a minimum post-Eocene dextral displacement of 100 km distributed along longitudinal faults. Paleomagnetic data in the Dauphinois zone [26,40] may also suggest counterclockwise rotation during the Tertiary and part of the total dextral displacement could have been accommodated west of the Penninic zone. Independent field evidence of post-Eocene dextral movements along major longitudinal faults are numerous in the Western Alps, especially along the Penninic front and other longitudinal faults of the Penninic and Dauphinois zones [5,10,15]. Field data of total displacement, based for example on displaced units, remain scarce due to the lack of similar paleogeographical units on both sides of the Penninic front. We may, however, note that the late Mesozoic–early Cenozoic Niesen flyschs of the Prealps show a heavy metal composition close to the Quermoz Valais flyschs, suggesting that these two basins were close together during their formation [41]. If true, a minimum of 80 km dextral displacement along the Penninic front is necessary to displace the Quermoz flyschs at its present-day location [6], in agreement with displacement deduced from paleomagnetic rotations.

Paleomagnetic data in the internal Alps are still too scarce to differentiate between the various possible mechanisms responsible for rotations, i.e. southwestward lateral expulsion of the western internal Alps along curved faults or counterclockwise rotation of the Apulia during the Cenozoic. However, the large rotations measured in the Durancian segment suggests that rotations have occurred at a larger scale and have probably affected the entire internal units of the western Alpine Arc. Indeed, within this domain, major transverse structures antithetic to the longitudinal faults have so far not been observed [15].

8. Conclusions

We present the first reliable paleomagnetic data on the Briançonnais sedimentary cover nappes of the western Alpine Arc. Three components of magnetization have been identified on Late Jurassic rocks of the Durancian segment of the Briançonnais zone: (1) a component of low unblocking temperature caused by a recent viscous overprint; (2) a second component (the A component) with intermediate unblocking temperature; (3) a component (the B component) of high unblocking temperature is present but poorly defined. The B component identification may be possible in the future by acquisition of new data. The A component is inferred to be a secondary magnetization acquired during Alpine post-metamorphic cooling. The direction of the A component implies that, since the Eocene, the study area of the Briançonnais zone has experienced a counterclockwise rotation of about 45º with respect to stable Europe. This result is in agreement with several kinematic models of the Western Alps [5–7] involving translations and counterclockwise rotations of the Penninic Alps during

the Tertiary. Rotations about vertical axes therefore appear to be an important mechanism accommodating the Tertiary deformation in the Penninic Alps. These rotations should be taken into account in any tectonic analysis of the pre-Oligocene kinematics of the Western Alps and in paleogeographic reconstructions of the Mesozoic alpine domain.

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